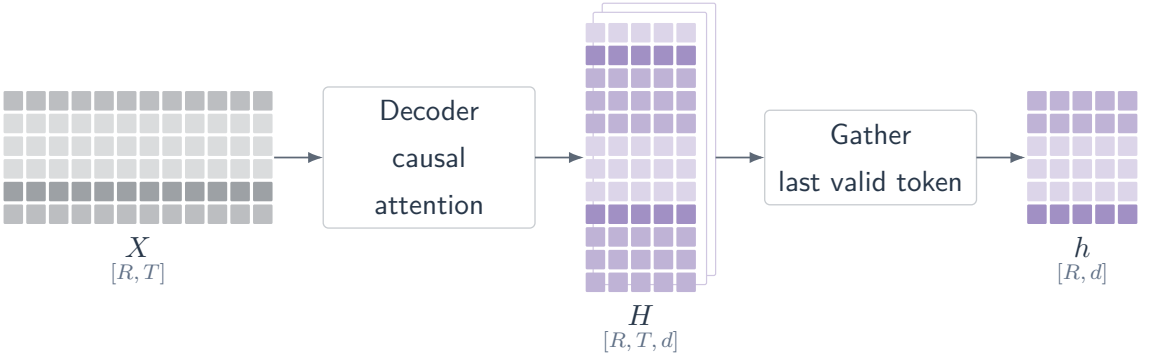
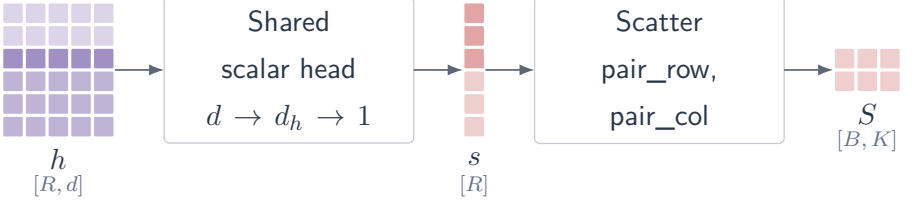
$$R = \sum_b K_b, \quad s_r = f_\theta(H_{r,-1,:}), \quad S_{\mathrm{row}(r),\mathrm{col}(r)} = s_r$$

One independent sequence per valid question–candidate pair



Score each row, then scatter back to question and candidate indices



Axes $R = \sum_b K_b$ counts valid pairs; $R = BK$ only when all questions have the same K. Rows are batchable, rather than K obligatory serial model calls.

Meaning A row contains the shared state/question and one candidate. Rows do not exchange hidden states. Softmax still couples the final probabilities within a question; runtime depends on batching, lengths and hardware.

Cell counts illustrate axes, not model sizes; tint identifies roles, not numeric values.